Hari Krishna Shah

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Summary

Full-stack software engineer with over 2 years of production experience building scalable web and geospatial systems, alongside an M.Tech in Computer Science at NIT Rourkela. Experienced in Node.js, React, PostgreSQL/PostGIS, real-time systems, and LLM-powered applications.

Education

National Institute of Technology, Rourkela — **Aug 2025 – Aug 2027**
M.Tech in Computer Science and Engineering — *Rourkela, Odisha, India*

Vellore Institute of Technology — **Aug 2021 – May 2024**
B.Sc. Computer Science | CGPA: 9.73/10 | Gold Medalist — *Vellore, Tamil Nadu, India*

Experience

Diginirman — **June 2024 – June 2026**
Lead Full Stack Developer — *Kathmandu, Nepal*

- Architected **Palika GIS**, a multi-tenant municipal GIS platform serving **25+ municipalities**, mapping **20,000+ km of roads** and **1M+ land and houses**.
- Built a real-time **Emergency Management System** on Socket.io: field reports (fire, road damage) stream live GPS location, severity, and photos/videos to a central dashboard for immediate response.
- Designed a **House Numbering System** that uses proximity algorithms on main and connecting roads to generate metric number plates, and a **Road Visualization System** storing type, length, width, drainage, and pavement metadata.
- Improved dashboard performance by caching computed statistics and spatial filters in **PostGIS**; built a **Digital Survey** module with live logging and batch processing for **100K+ responses**.
- Delivered a WebSocket-based bulk download service, automated Dropbox backups with status monitoring, and multi-user land-zone recommendation with live commenting.

Projects

Palika GIS – AI Chatbot & Digital Profile — palikagis.com

- **NL-to-SQL Chatbot**: Built a LangChain pipeline over a **150+ table** schema. The LLM receives the query, chat history, and schema, and generates SQL; the system executes it and returns the result to the LLM to produce a natural-language answer.
- **Digital Profile**: Filtering across **50+ demographic attributes** (language, disease, vehicle, etc.) to generate ward- and tole-level reports. *Tech:* React, Node.js, PostgreSQL, PostGIS, Socket.io, Leaflet.js, LangChain.

Face Recognition System and Incident Reporting System, Rourkela Police — raurkelapolice.in

- Contributed to a ResNet-50 based face recognition system to support identification workflows. *Tech:* Python, ResNet-50, Computer Vision. Also, developed live incident reporting system for citizens.

Tetra Labs – Civil Testing Lab SaaS — civilelab.com

- Automated the workflow from test request to lab data entry, report generation, and a stakeholder download portal with **RBAC**. *Tech:* Next.js, TypeScript, PostgreSQL, GraphQL.

Linfra Go – Geospatial Visualization Platform

- Web GIS for orthomosaics, 3D point clouds, contours, DEMs, and DTMs; tiles large point-cloud datasets (**100 GB+**) for responsive in-browser rendering. *Tech:* Next.js, Leaflet, Potree, GDAL, MBTiles.

Municipal Portal — ward10.lmc.gov.np

- Modular CMS, events, and employee directory backend adopted by multiple municipalities. *Tech:* React, Node.js, Express, Postgres SQL.

Technical Skills

Languages: TypeScript, JavaScript, Python, C, C++, SQL
Frontend: React, Next.js, Redux, Tailwind CSS
Backend: Node.js, Express, GraphQL, REST, Socket.io, WebSockets
Databases: PostgreSQL, PostGIS, MongoDB, MySQL
AI/ML: LangChain, LLM applications, NL-to-SQL, CNNs, Computer Vision
Tools: Git, Docker, AWS, DigitalOcean, Linux, Leaflet, GDAL
Core CS: Data Structures, Algorithms, System Design, DBMS, Computer Networks

Certifications & Publication

MERN Stack Development – 135 hours, Broadway Infosys — Verify
Front-End Web Development – 90 hours, Broadway Infosys — Verify
C & C++ Training – Broadway Infosys — Verify
Paper: *Comparison of Optimization Techniques for Detection of Fruit Diseases*, ICSCDS-2023.